

LACHOUB HANANE

Computer Science Engineer
Edge AI and Assistive Technology

+213 553 768 349 • hananelachoub03@gmail.com • Tlemcen, Algeria
Lachoub Hanane • github.com/hananiiii

Research Profile

Computer Science Engineer with research experience in efficient computer vision, Edge AI, and multimodal sensing. First and corresponding author of a manuscript under review on NoorTech, an offline-first assistive wearable using lightweight YOLOv5n inference on Raspberry Pi, achieving 80.9% mAP@0.5 across 15 outdoor navigation classes. Released a public dataset of approximately 17,000 images and developed multimodal sensor fusion for real-time assistance. Interested in resource-efficient AI computing, on-device inference, and reliable computer vision systems under computational constraints.

Research Interests

• Resource-Efficient AI Computing • EdgeAI • Efficient Computer Vision • Model Optimization and Deployment • Multimodal Sensor Fusion • • Real-Time Intelligent Systems • Edge–Cloud Collaboration and Computation Offloading

Education

Master's and Engineering Degree in Computer Science July 2026
Higher School of Computer Science (ESI-SBA), Algeria 2021–2026

GPA: **3.85/4.00** (**15.07/20** on the Algerian scale)
Integrated five-year program: preparatory classes (2021–2023), engineering cycle (2023–2026)

Scientific Baccalaureate June 2021
Bouazza Miloud High School, Tlemcen, Algeria 2018–2021

Publications & Research Outputs

- **Hanane Lachoub**, Manel Baba Ahmed, Abdelatif Rahmoun, NoorTech: An AI-Powered Wearable Device for Visually Impaired Persons. Manuscript submitted under review, 2026. First author and corresponding author.
- Lachoub, H. Assistive Belt Outdoor Navigation Dataset for Visually Impaired: 15-Class Object Detection Dataset, ∼17,000 images, ∼1,500 instances per class, YOLO format. Lachoub, H. (2026). Version 1.0.0 [Dataset]. Zenodo. [DOI]

Research Experience & Theses

NoorTech Belt — AI-Powered Assistive Intelligent System Engineering Thesis – Final Year Project

- Designed and developed a multimodal assistive wearable for visually impaired users, integrating camera, ultrasonic sensors, GPS/GSM, and speech technologies on a Raspberry Pi for real-time assistance.
- Trained and deployed a lightweight YOLOv5n model (1.9M parameters, 3.8 MB ONNX) for CPU-based edge inference, achieving **80.9% mAP@0.5** across 15 outdoor navigation classes after comparing it with YOLOv11n.
- Designed multimodal sensor fusion combining camera and ultrasonic measurements with detection confidence, distance, direction, urgency, and contextual information; integrated Vosk, Piper, and Tesseract for fully local speech and OCR processing.
- Designed a context-aware human–AI interaction framework enabling the system to **warn, act, wait, or request confirmation** according to uncertainty, user interaction, and potential consequences.
- Built and publicly released a custom dataset of approximately **17,000 images** covering **15 outdoor navigation classes**, with approximately 40% manually collected and annotated images.

[Thesis] [GitHub]

Edge AI for Assistive Technologies Master's Thesis

- Conducted a literature review of **12 Edge AI-based assistive systems** published between 2019 and 2026, analyzing sensing modalities, system architectures, and local versus cloud processing.
- Identified deployment challenges for resource-constrained devices, including limited computation, battery consumption, connectivity dependence, device fragmentation, and real-time processing requirements.
- Synthesized requirements for reliable, human-centered Edge AI, emphasizing multimodal sensing, local inference, contextual adaptation, and resource-aware deployment.

[Thesis]

Awards and Leadership

- **ProtoMarket II — National Innovation Challenge Acceleration Program | 2026** — Selected as a winning project for NoorTech, an AI-powered assistive wearable for people with visual impairments, among 2,110 applications; awarded 1,000,000 DZD in project funding. [Link]
- **INNOVPOST Hackathon — 1st Place, AI Computer Vision Track | 2024** — Led Team NeuralDetect in developing an AI-powered damaged-package detection system, winning first place in the track at Algérie Poste's national hackathon. [Link] [GitHub]
- **Team Lead, KhidmaTic** — ESI-SBA. Led a team developing a home-maintenance platform, coordinating system architecture, task allocation, and project progress. [GitHub]

Selected AI & Research Projects

Agentic Medical Question-Answering System AI / Intelligent Systems Project

Developed a LangGraph-based medical QA system with an LLM router selecting among medical databases, device manuals, and web search, with relevance grading and adaptive retries. [GitHub]

Medical Image Classification Computer Vision Project

Developed transfer-learning models for blood cancer cell classification using **EfficientNetB3** and retinal disease classification using **VGG19** across four classes. [Blood Cell GitHub] [Retinal GitHub]

Professional & Teaching Experience

- **Freelance Software Engineer** — Algeria, Remote. Developed software solutions from requirements analysis and architecture to implementation, including backend services, REST APIs, databases, and authentication using Python and JavaScript/TypeScript.
- **Software Engineering Intern** — Information Systems Department, Industrial Company. Analyzed cross-department information flows and documented functional requirements to support workflow and information-system design.
- **Korean Language Instructor** — Independent & Tech_Informatique Language School. Designed and delivered online Korean language courses for international learners at different proficiency levels.

Technical Skills

Edge AI & Deployment: Raspberry Pi, ONNX, On-Device Inference, Offline AI, CPU-Based Inference, Real-Time Processing

Computer Vision & Machine Learning: Python, PyTorch, YOLO, CNNs, Transfer Learning, Image Segmentation, OpenCV, Scikit-learn, Pandas, NumPy,

Multimodal Systems: Sensor Fusion, OCR, Speech-to-Text, Text-to-Speech, Context-Aware Intelligent Systems

Human–AI Interaction & Intelligent Systems: Human–AI Interaction, Human-Centered AI, RAG, LangChain, LangGraph

Software Engineering: JavaScript, TypeScript, React, Next.js, Node.js, FastAPI, MongoDB, MySQL, PostgreSQL, Docker, Git

Languages

Arabic — Native | **English** — Advanced | **French** — Intermediate | **Korean** — Intermediate (certificat level 4B)

References

Prof. Abdellatif Rahmoun, Higher School of Computer Science (ESI-SBA), LabRI-SBA Lab, rahmoun@esi-sba.dz [Google Scholar Link] [ORCID Link]

Dr. Manel Baba-Ahmed, Higher School of Computer Science (ESI-SBA), LabRI-SBA Lab, m.baba-ahmed@esi-sba.dz